Northwind Catalog Application - Complete Documentation

1. Introduction

This project is a full-stack application developed using ASP.NET Core. It uses a layered architecture with Web (MVC), Services (API), and Database (SQL Server).

2. Project Setup Steps

1. Created solution in Visual Studio
2. Added Services project (ASP.NET Core Web API)
3. Added Web project (ASP.NET MVC)
4. Connected both projects using HttpClient
5. Configured SQL Server Northwind database
6. Used Scaffold-DbContext to generate models

3. Database Configuration

Connection string added in appsettings.json. Used Entity Framework Core (DB First approach). Tables used: Categories, Products. Relationship: One Category has many Products.

4. Models Generation

Used command: dotnet ef dbcontext scaffold ... --table Products --table Categories Generated Product, Category and DbContext classes.

5. DTO Layer

Created DTOs:

- CategoryDto
- ProductDto
- CategorySummaryDto Used to transfer data between API and UI.

6. Repository Pattern

Implemented repository pattern to separate data access logic. Created interfaces and implementations for Category and Product.

7. API Development

Created API Controllers:

- CategoriesApiController
- ProductsApiController Endpoints:
- GET /api/categories
- GET /api/products/by-category/{id}
- GET /api/products/summary

8. MVC Integration

Used HttpClient in MVC to call API. Created controllers:

- CategoriesController
- ProductsController
- SummaryController Views created for displaying data.

9. UI Implementation

Categories shown as cards. Products shown in table/list with prices. Summary shown with analytics (count, avg price, etc.).

10. Unit Testing (xUnit)

Created test project using xUnit. Tested InventoryValue logic. Used Arrange-Act-Assert pattern. Example: UnitPrice = 10, UnitsInStock = 5 $\rightarrow$ InventoryValue = 50.

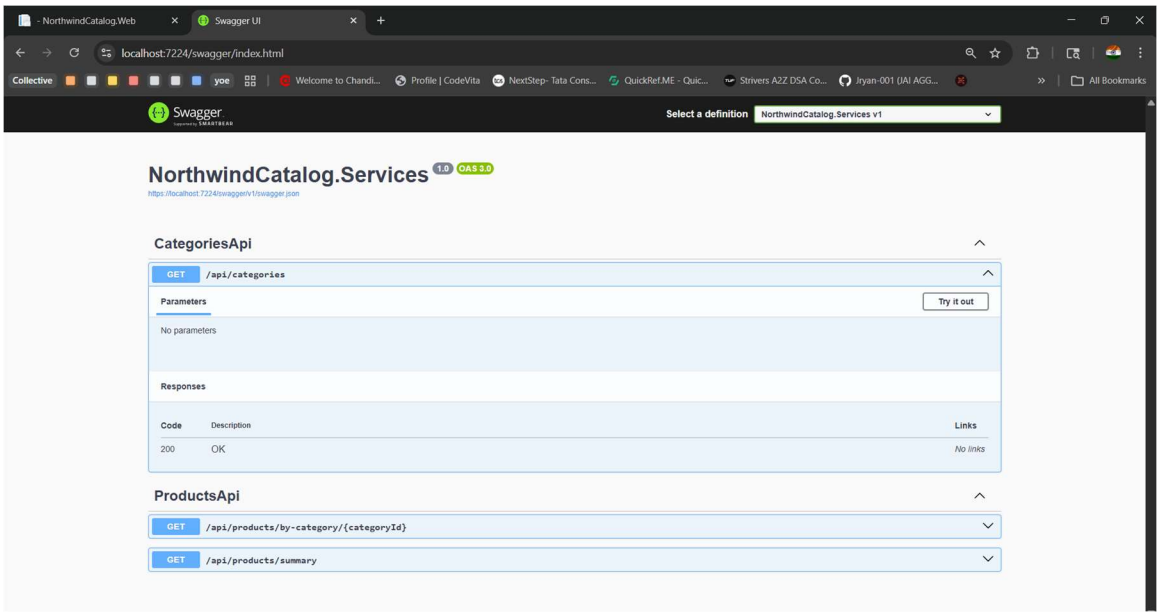
11. Running the Project

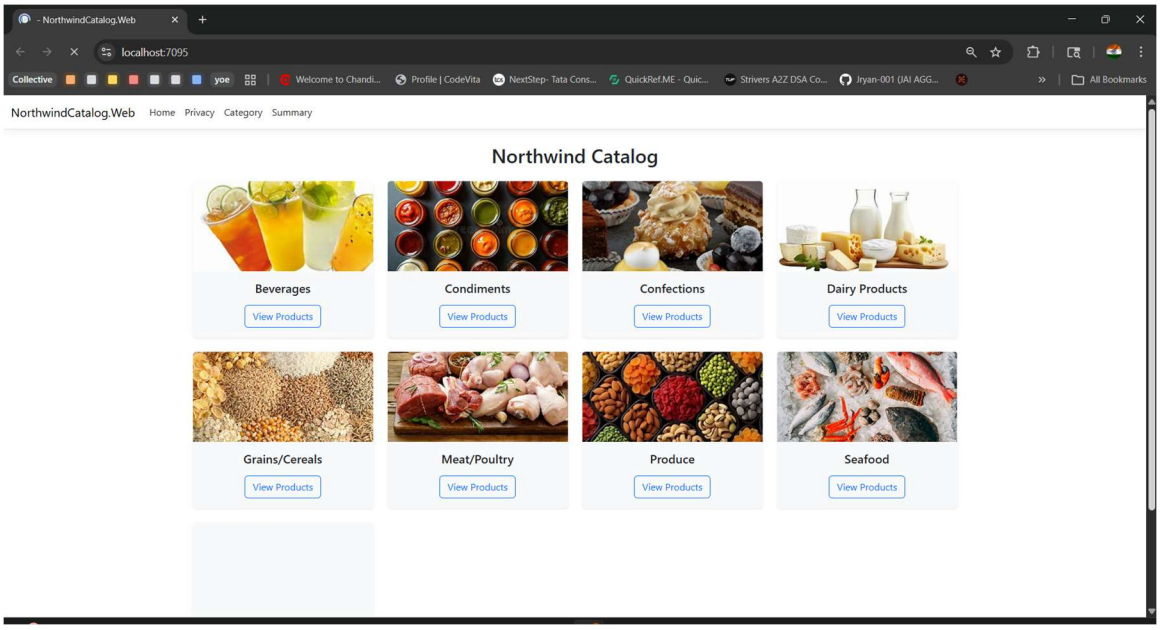
Step 1: Run Services (API) Step 2: Run Web (MVC) Step 3: Open /Categories Step 4: Navigate to Products and Summary pages.

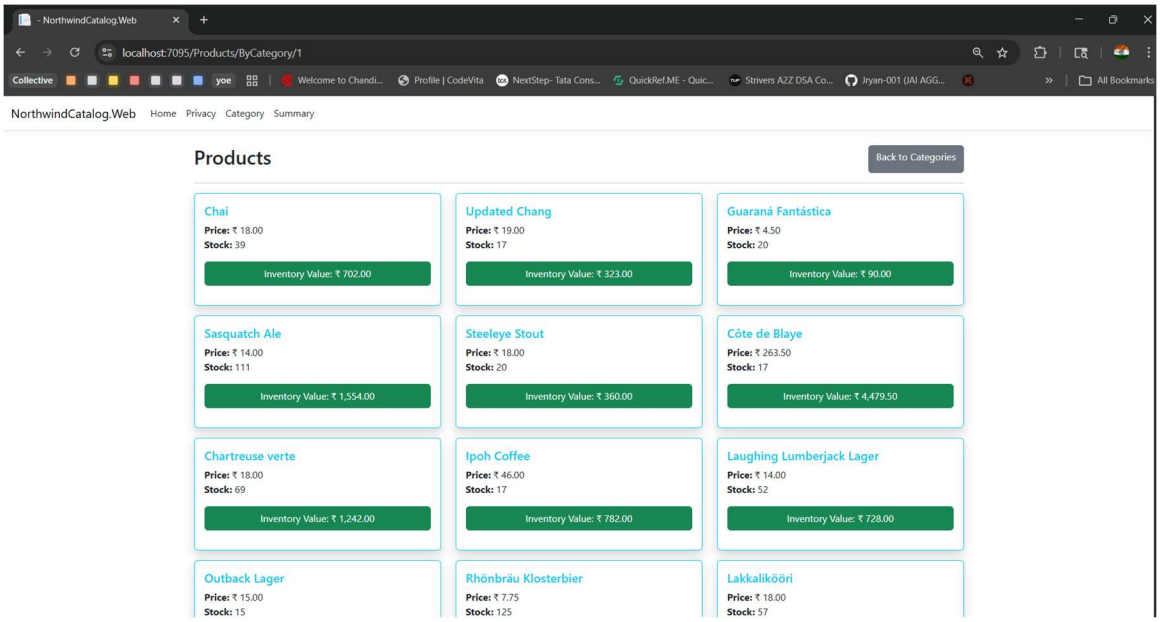
12. Conclusion

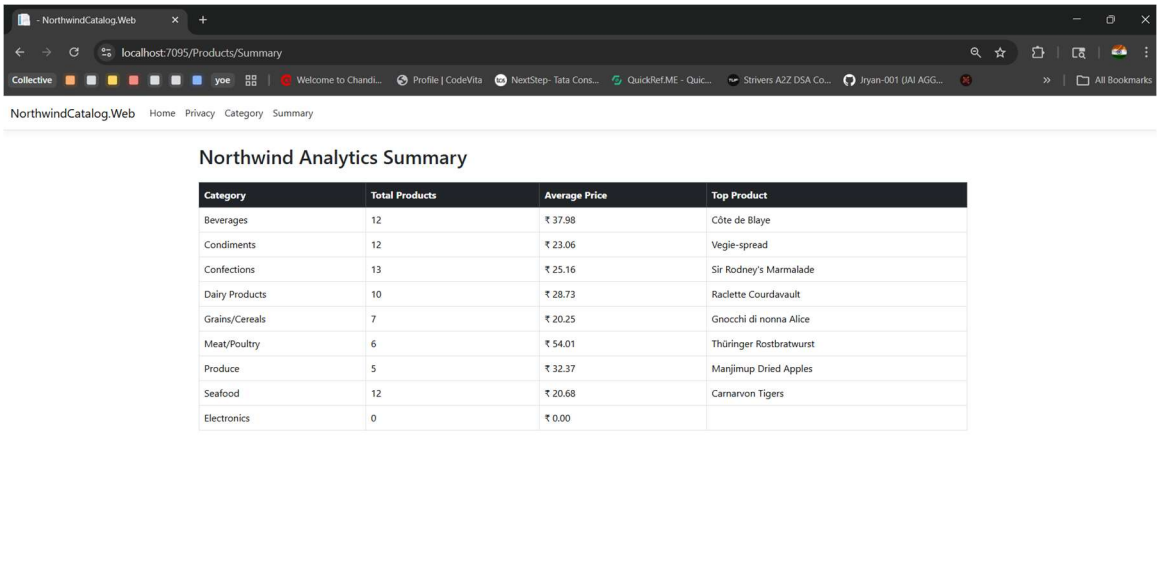
Successfully developed and tested a full-stack application with API, MVC frontend, and unit testing.

13. Output Screenshots









Category	Total Products	Average Price	Top Product
Beverages	12	₹ 37.98	Côte de Blaye
Condiments	12	₹ 23.06	Vegie-spread
Confections	13	₹ 25.16	Sir Rodney's Marmalade
Dairy Products	10	₹ 28.73	Raclette Courdavault
Grains/Cereals	7	₹ 20.25	Gnocchi di nonna Alice
Meat/Poultry	6	₹ 54.01	Thüringer Rostbratwurst
Produce	5	₹ 32.37	Manjimup Dried Apples
Seafood	12	₹ 20.68	Carnarvon Tigers
Electronics	0	₹ 0.00	

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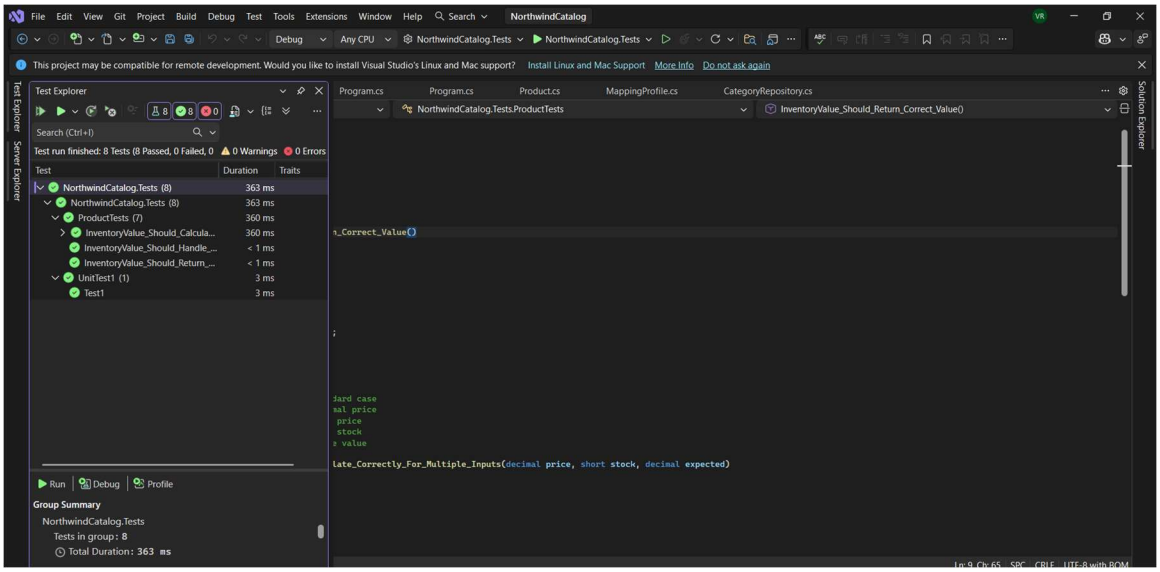
File Edit View Git Project Build Debug Test Tools Extensions Window Help Search NorthwindCatalog
Debug Any CPU NorthwindCatalog.Tests NorthwindCatalog.Tests
This project may be compatible for remote development. Would you like to install Visual Studio's Linux and Mac support? Install Linux and Mac Support More Info Do not ask again
Test Explorer
Search (Ctrl+I)
Test run finished: 8 Tests (8 Passed, 0 Failed, 0 0 Warnings 0 Errors
Test Duration Traits
NorthwindCatalog.Tests (8) 363 ms
NorthwindCatalog.Tests (8) 363 ms
ProductTests (7) 360 ms
InventoryValue_Should_Calcula... 360 ms
InventoryValue_Should_Handle_... < 1 ms
InventoryValue_Should_Return_... < 1 ms
UnitTest1 (1) 3 ms
Test1 3 ms
Run Debug Profile
Group Summary
NorthwindCatalog.Tests
Tests in group: 8
Total Duration: 363 ms
Program.cs Program.cs Product.cs MappingProfile.cs CategoryRepository.cs
NorthwindCatalog.Tests.ProductTests
InventoryValue_Should_Return_Correct_Value()
Test Explorer Server Explorer
Solution Explorer